Retail Chain Profit Leakage & Demand Optimization

Project Summary

What I worked on

The aim was to understand why profit can fall even when sales are strong. I cleaned and analysed sales, products, inventory, returns, vendors, promotions and customer data using SQL, Python, Power BI and Tableau.

1. Overall numbers

Metric	Result
Total Sales	Rs. 26.47M
Gross Profit	Rs. 2.96M
Net Profit	Rs. 1.46M
Approved Return Loss	Rs. 1.51M
Potential Stockout Lost Sales	Rs. 3,059.58 (11 combinations; 25 units of recorded demand)

2. Main profit leakage found

The five highest-loss SKUs in the updated Tableau view were UrbanLeaf Rice Combo (about -Rs. 11.29K), UrbanLeaf Ice Cream 1L (about -Rs. 9.94K), HealthHub Diapers Pack (about -Rs. 8.08K), CleanPro Curd 1kg (about -Rs. 7.52K), and Snackify Rice Combo (about -Rs. 6.03K). Together, these represent approximately Rs. 42.85K of negative net profit. The ranking highlights products for review of price, discounts, product cost and refunds.

3. Returns and promotions

Groceries had the highest return contribution in the category analysis, followed by Dairy, Home Care and Frozen. Promotion analysis compared incremental profit with a non-promotion baseline across 195 promotion records. Some promotions showed negative incremental profit, meaning extra volume did not translate into extra profit. A discount rate above 10% was used as a working review threshold because no fixed threshold was specified.

4. Vendor and stockout findings

Vendor performance was assessed using late purchase orders, average delay and SLA breach percentage. VEND0026, VEND0018 and VEND0001 had the highest late-PO counts among the vendors reviewed.

Updated stockout result: the refreshed SQL/Tableau analysis identified 11 SKU-store-date combinations with zero closing stock and matching delivered demand. Combined recorded demand was 25 units, with estimated potential lost sales of **Rs. 3,059.58**. This is an estimate based on recorded demand during zero-stock periods, not proof of all sales that could have been lost. Vendor delays are treated as a separate risk unless a matching supply record supports a causal link.

5. Customer segmentation

Segment	Customers	Share
Low-Engagement Customers	6,328	42.47%
High-Value Loyal Customers	3,759	25.23%
High-Value High-Return Customers	3,082	20.68%
High-Ticket Customers	1,732	11.62%

Segmentation used total spend, average order value, order frequency and return frequency. K-Means clustering with four groups was selected after comparing a few cluster sizes.

6. What I built

Excel: cleaned data and data-quality report
SQL Server: database tables, analysis queries and reporting views
Python: validation, customer segmentation and automated insight tables/reports
Power BI: operational dashboard with KPIs and detailed views
Tableau: executive summary, store performance quadrant, promotion analysis and vendor/inventory view

Demand forecasting was optional in the project instructions, so a forecasting model was not included.

7. Practical next steps

1. Review the five loss-making SKUs and their selling price, discounts, unit cost and refunds.
2. Investigate categories with higher return contribution and understand return reasons.
3. Review vendors with repeated SLA breaches.
4. Review promotions with negative incremental profit.
5. Monitor products and stores where recorded demand occurs during zero-stock periods.

Conclusion

High sales do not automatically mean high profit. Returns, discounts, product cost, promotions and stock availability all affect profitability. The refreshed dashboards and analyses help prioritize where the business should investigate first.